

Matthijn Dijkstra

STAFF ENGINEER · ARCHITECT

Amsterdam, Netherlands · Remote or hybrid · Available for customer travel

Dutch citizen · Dutch & English · CET, with US overlap

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PROFILE

Software engineer and architect who joined PeerWell as its first full-time engineer, built much of the initial digital therapeutics platform, and led system design as the engineering team grew. Later led its integration into Bardavon and relaunch as Recovery+. Hands-on across mobile, web, backend, infrastructure and applied ML, with responsibility for technical direction, production delivery and engineering handover. Earlier independent client work covered scoping, implementation, deployment and ongoing support.

EXPERIENCE

Bardavon Health Innovations

2022 - 2025 · Remote

Lead Engineer, Acquisition Integration

- Determined what to retain or rebuild as PeerWell shifted from surgical preparation and recovery to injured-worker care. Launched Recovery+ in April 2024, connecting injured workers with licensed clinical coaches.
- Migrated the platform onto Bardavon's infrastructure using AWS Fargate, Docker, GitHub Actions and Terraform, and integrated it with their visual workflow automation platform.
- Reconciled engineering practices and assumptions about the codebase across the merged teams.
- Transferred full platform ownership to Bardavon's engineering team through the migration, shared implementation work and architecture handover. Completed in 2025.

PeerWell

2016 - 2022 · San Francisco / Remote

First Engineer to Engineering Lead

- Designed and built most of the initial digital therapeutics platform for surgical preparation and recovery, used by patients and incorporated into clinical care pathways. Built the patient data model and applications with React, Node/TypeScript and DocumentDB, supported by HIPAA-compliant infrastructure.
- Designed an adaptive recommendation engine that personalises recovery pathways using patient context, historical outcomes and configurable clinical scoring. Its architecture is described in published U.S. patent application [US20180286509A1](#) (co-inventor).
- Built image and pose recognition for AI-assisted movement tracking, plus an authoring tool that let the content team define exercises from sample videos.
- Led system design as the team grew across the US and Europe. Mentored engineers, ran interviews and helped set the hiring bar; stayed through funding rounds and acquisition.

Published platform outcomes: 26% shorter hospital stays and an 80% relative increase in home discharge (peer-reviewed feasibility study, n=476, *Annals of Translational Medicine*, 2019;7(4)). A separate Bardavon case study reported 35% average pain reduction over a 10-week program (ODI-measured).

Freelance

2007 - 2016 · Netherlands

Independent Software Engineer

Nearly nine years of client work alongside study and later full-time employment. Usually the sole engineer, working with designers and the client's own team.

- Owned the full delivery loop: clarified what the client needed, agreed deliverables, built and deployed the solution, and supported it in operation.
- Built web applications and integrations for clients across different businesses, adapting the technical approach to their needs and existing systems.

SELECTED WORK

Nabu

2025 - present · Open source, solo

AI-native Integrated Research Environment

A local-first research environment that turns source documents into structured information people can query and explore.

- Combines LLM-powered qualitative coding, RAG with HyDE, structured data extraction, multi-model consensus voting, SQL querying and document versioning.
- Keeps prose documents as the source of truth, with incrementally maintained projections into DuckDB-WASM for querying and memoized selectors for UI state.

github.com/mdijkstra-oss/nabu-frontend

SKILLS

LANGUAGES	Go, TypeScript/JavaScript, Swift, Python, SQL
FRONTEND	React, TypeScript, native iOS (Swift), Storybook, internationalisation, accessibility, offline and local-first applications
BACKEND	Node, REST APIs, microservices, SQL and NoSQL
INFRASTRUCTURE	AWS, Fargate, Docker, Terraform, GitHub Actions, CI/CD, observability
AI / ML	LLM pipelines and agentic systems, RAG, multi-model consensus and evaluation, computer vision and pose estimation
PRACTICE	System design, architecture documentation and handover, test strategy and CI quality gates, HIPAA-regulated environments
LEADERSHIP	Mentoring and technical leadership, hiring and interviewing, cross-team integration after acquisition
SPOKEN	Dutch, English

EDUCATION

Hanze University of Applied Sciences · BSc Computer Science · 2011 - 2016

ROC · Computer and Network Systems, CCNA · 2005 - 2009